

path length $r_0 \sqrt{1 + 4P^2} = 2.35 r_0 > 2r_0$

$\Gamma_1 = 1$

$\Gamma_2 = 1/2$

$\Gamma_3 = -1/3$

r_0

$\arctan 2P = 64.8^\circ > 60^\circ$

Diagram illustrating a path length calculation. A solid arc connects two points, $\Gamma_1 = 1$ and $\Gamma_2 = 0.05$, with a radius r_0 . The arc length is given as $\text{path length } r_0 \sqrt{1 + 4P^2} = 2.005 r_0 > 2r_0$. The angle subtended by the arc is $\arctan 2P = 60.1^\circ > 60^\circ$. The value $\Gamma_3 = -0.048$ is also indicated.